

MEMORANDUM: THE LOCAL FACTORS BEHIND THE TWIST-SYMMETRISED BOUND

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ABSTRACT. This memorandum answers the question raised by both referee reports on the note *Local component groups and the ℓ -adic valuations of the modular degree*: whether the twist-symmetrised local weights of Conjecture 1.3 are the local factors of the adjoint / congruence-ideal theory, and if not, what the exact local factors are. The answer is that the correct local term at a tame additive prime q is the ℓ -adic valuation of the inverse Euler factor of the symmetric square L -function of f_E at $s = 2$ (equivalently of the adjoint at $s = 1$), and that the twist-symmetrised weights were its shadow. The new rule is sharper than the one in the note, holds with no exception at $\ell \geq 5$ on every curve of conductor at most 500000 and on the 893 curves outside the tables, and at $\ell = 3$ fails only in a configuration of a precise shape (264 curves in 2.1 million), involving a prime of type III together with a potentially multiplicative prime $\equiv -1 \pmod{3}$ at which the residual representation of the twist is unramified, which is exactly the case excluded by the hypotheses of Kim and Ota; without the type III terms there is no exception at all. Parts of the new rule are theorems by the existing theory; the rest is a statement about the integrality of a primitive adjoint L -value. The memorandum ends with the comparison table the reports asked for and with a recommendation for the revision of the note.

1. SUMMARY

Let $E/\mathbb{Q}$ be the $X_0(N)$ -optimal curve of a non-CM isogeny class, $f = f_E$ its newform, ℓ an odd prime not dividing N with $E[\ell]$ irreducible, and r_E the congruence number, so that $v_\ell(\deg \varphi_E) = v_\ell(r_E)$ by Agashe–Ribet–Stein. For a prime $q \neq \ell$ let $V = V_\ell E$ and let

$$P_q(X) = \det(1 - \text{Frob}_q X \mid (\text{Sym}^2 V)^{I_q}),$$

so that $L_q(\text{Sym}^2 f, s) = P_q(q^{-s})^{-1}$ is the local factor of the symmetric square L -function, and define

$$(1) \quad e_\ell(E, q) = v_\ell\left(\frac{P_q(q^{-2})}{1 - a_q(E)^2 q^{-2}}\right),$$

the valuation of the ratio between the true inverse Euler factor at $s = 2$ and the “naive” one built from the coefficient a_q of f . For a multiplicative prime the two agree and $e_\ell(E, q) = 0$; for an additive prime $a_q = 0$ and $e_\ell(E, q) = v_\ell(P_q(q^{-2}))$. For $q \geq 5$ this is computed in Section 4 from the Kodaira type:

type at $q \geq 5$	$(\text{ad}^0 V)^{I_q}$	$e_\ell(E, q)$
I_n ($n \geq 1$)	line; $L_q(\text{Sym}^2 f, s) = (1 - q^{-s})^{-1}$	0 (the naive factor is the same)
I_n^*	line; $L_q(\text{Sym}^2 f, s) = (1 - q^{-s})^{-1}$	$v_\ell(q^2 - 1)$
$\text{II}, \text{II}^*, \text{IV}, \text{IV}^*$	line, Frobenius eigenvalue $\chi_{-3}(q)q$	$v_\ell(q - \chi_{-3}(q))$
III, III^*	line, Frobenius eigenvalue $\chi_{-4}(q)q$	$v_\ell(q - \chi_{-4}(q))$
I_0^*	all of $\text{ad}^0 V$ (unramified)	$v_\ell(q - 1) + v_\ell((q + 1)^2 - a_q(E')^2)$

where χ_{-3}, χ_{-4} are the quadratic characters of $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(i)$, the eigenvalues are those of the arithmetic Frobenius on the invariant line of $\text{Sym}^2 V$ in the classical normalisation in which the

unramified factor is $(1 - \alpha^2 X)(1 - \alpha\beta X)(1 - \beta^2 X)$, and E' is the quadratic twist of E by q^* , which has good reduction at q when E has type I_0^* . Let also $t_\ell(E, q) = v_\ell(n)$ for types I_n and I_n^* (q odd) and 0 otherwise, the Tamagawa exponent of $\text{ad}^0 V$ at q . The data say:

Conjecture 1.1 (Euler-factor rule). *With the hypotheses above, let the sum run over the primes $q \geq 5$, $q \neq \ell$, of types other than III, III* for the term e_ℓ , and over all primes for the term t_ℓ (the primes 2 and 3 entering only through t_ℓ and the empirical weights of Section 5). Then*

$$(2) \quad v_\ell(\deg \varphi_E) \geq \sum_{q|N} (e_\ell(E, q) + t_\ell(E, q)).$$

Moreover the term $e_\ell(E, q) = v_\ell(q - \chi_{-4}(q))$ of a prime of type III or III* can be added as well, except possibly when some prime $p \equiv -1 \pmod{\ell}$ of type I_n or I_n^* has $\ell \mid n_p$.

Every term of (2) is attained as a minimum, in isolation, over tens of thousands of curves (Section 5); (2) holds with no exception on all four ranges at $\ell = 3, 5, 7, 11, 13$; with the III terms added it has 264 exceptions at $\ell = 3$ and none at $\ell \geq 5$, all of the shape described in Section 5. It implies the two conjectures of the note: $3 \mid q - \chi_{-3}(q)$ for every $q \geq 5$, which is why types II and IV always force a factor 3, and $3 \mid q^2 - 1$, which is why type I_n^* always does, whatever n . The twist-symmetrised weights are explained: $\text{ad}^0 V$ does not see a quadratic twist, so neither does (1).

2. THE FRAMEWORK

We use the Σ -ramified deformation rings and Hecke algebras of Diamond–Flach–Guo and Dimitrov in the formulation of Kim and Ota [4, §2]: for $\bar{\rho} = \bar{\rho}_{E, \ell}$ and a finite set Σ of primes other than ℓ , T_Σ is the Hecke algebra acting on the Σ -ramified deformations, which live at level $N_{\bar{\rho}, \Sigma} = r \cdot N(\bar{\rho}) \prod_{p \in \Sigma} p^{\dim \bar{\rho}^{I_p}}$ (an auxiliary prime r aside), and the congruence ideal of a newform f in T_Σ is $\eta_{f, \Sigma} = \pi_f(\text{Ann ker } \pi_f)$. For an elliptic curve and Σ the set of primes dividing $N/N(\bar{\rho})$, $\eta_{f, \Sigma}$ is the ℓ -part of the congruence number r_E [4, Lemma 2.7]. Three results are used.

$R = T$ and the adjoint Selmer group [4, Theorem 2.9], from [2, 3]: under (FL) $2 \leq k \leq \ell - 1$, $\ell \nmid N$, and (TW) $\bar{\rho}|_{\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))}$ absolutely irreducible, $R_\Sigma = T_\Sigma$ and $\#\text{Sel}_\Sigma(\mathbb{Q}, \text{ad}^0 f \otimes E/\mathcal{O}) = \#\mathcal{O}/\eta_{f, \Sigma}$ for every newform f in T_Σ . Note that $\text{ad}^0(\rho \otimes \chi) = \text{ad}^0 \rho$ for any character χ ; Kim and Ota remark (their Remark 2.2, after [2, §1.7.1]) that one may therefore assume the newforms to have minimal conductor among their twists. This is the structural reason for the twist-symmetrised weights of the note.

Hida's formula for the congruence ideal [1, Theorem 4.20]: for f in T_Σ , with $N = N_\Sigma$ and N_f the level of f ,

$$(3) \quad \eta_{f, \Sigma} = \left(\frac{N_f L(\text{Sym}^2 f, 2)}{i\pi \det A} \right) \prod_{p|N/N_f} c_p(f), \quad c_p(f) = p^{\delta_p} L_p(\text{Sym}^2 f, 2)^{-1} \ (p \neq \ell),$$

where $L(\text{Sym}^2 f, s)$ is the naive symmetric square built from the Euler product of $\sum a_n(f)^2 n^{-s}$ (so with trivial factor at primes $p^2 \mid N_f$, where $a_p = 0$) and $\det A$ is a period of the lattice $H_1(X_0(N), \mathbb{Z})[\wp]$. The factors c_p for $p \in \Sigma$ with $p \mid N/N_f$ are inverse Euler factors of $\text{Sym}^2 f$ at $s = 2$: this is Wiles's change-of- Σ formula, and it is where the terms e_ℓ of (1) come from when the level is raised at p .

Quantitative level lowering [4, Theorem 1.1]: for $N = N^+ N^-$ with N^- square-free, $\ell \geq 5$, $\ell \nmid N$, (TW), and (6): $\bar{\rho}$ ramified at every $q \mid N^-$ with $q \equiv \pm 1 \pmod{\ell}$,

$$(4) \quad \text{ord}_\lambda \eta_f(N) = \text{ord}_\lambda \eta_f(N^+, N^-) + \sum_{q|N^-} t_f(q),$$

with $\eta_f(N)$ the congruence ideal in the full space $S_2(\Gamma_0(N))$, $\eta_f(N^+, N^-)$ the one in the N^- -new subspace, and $t_f(q)$ the Tamagawa exponent, equal to $v_\ell(n_q)$ for an elliptic curve. Condition (6) is exactly where the Euler factor $1 - q^{-2}$ of a Steinberg prime is a non-unit; the authors need it to control the “unit Euler factors” in their L -value computation (their Proposition 2.15).

3. TWIST MONOTONICITY OF THE CONGRUENCE NUMBER

The following elementary statement transports the multiplicative theory to the twisted types and explains the equality of the contributions of the types exchanged by a twist.

Theorem 3.1. *Let $E/\mathbb{Q}$ have conductor N , let S be a set of odd primes, $D = \prod_{q \in S} q$, χ_S the product of the quadratic characters of conductor q , $q \in S$, and $E_S = E \otimes \chi_S$ of conductor N_S . If $q^2 \mid N$ for every $q \in S$ and $\text{lcm}(N_S, D^2) \mid N$, then r_{E_S} divides r_E . If moreover $\text{lcm}(N, D^2) \mid N_S$, then $r_{E_S} = r_E$.*

Proof. Let $f = f_E$, $g = f_{E_S}$, and $\iota: S_2(\Gamma_0(N_S)) \rightarrow S_2(\Gamma_0(\text{lcm}(N_S, D^2))) \subset S_2(\Gamma_0(N))$, $h \mapsto h \otimes \chi_S = \sum \chi_S(n) a_n(h) q^n$. The map ι preserves integrality of coefficients, sends eigenforms to eigenforms (twisting the eigenvalues at $p \nmid D$ by $\chi_S(p)$, killing the coefficients at multiples of D), and sends g to f : both are multiplicative, $a_p(f) = \chi_S(p) a_p(g)$ for $p \nmid D$ because the twist is unramified outside S , and $a_{q^k}(f) = 0 = \chi_S(q^k) a_{q^k}(g)$ for $q \in S$ because $q^2 \mid N$. An eigenform $h \neq g$ of level dividing N_S has an eigensystem different from that of g at infinitely many primes, so $\iota(h)$ has an eigensystem different from that of f and lies in $f^\perp$; hence $\iota(g^\perp \cap S_2(\Gamma_0(N_S); \mathbb{Z})) \subset f^\perp \cap S_2(\Gamma_0(N); \mathbb{Z})$. If $g \equiv h \pmod{r}$ with $h \in g^\perp$ integral, then $f = \iota(g) \equiv \iota(h) \pmod{r}$ with $\iota(h) \in f^\perp$ integral, and $r \mid r_E$ by the cyclicity of the congruence module (Lemma 3.1 of the note). Taking $r = r_{E_S}$ gives $r_{E_S} \mid r_E$; the second hypothesis makes the situation symmetric. $\square$

The hypotheses hold in three cases. (a) E has type I_n^* at each $q \in S$: then E_S has type I_n at q , $N_S = N/D$, and $\text{lcm}(N_S, D^2) = N$; so $r_{E_S} \mid r_E$. (b) E has type I_0^* at each $q \in S$: E_S has good reduction at q , $N_S = N/D^2$, again $r_{E_S} \mid r_E$. (c) E has type II, III, IV or a dual at each $q \in S$, $q \geq 5$: then $N_S = N$ and $r_{E_S} = r_E$; so for every prime p with $p^2 \nmid N$, $v_p(\deg \varphi_E) = v_p(\deg \varphi_{E_S})$, and the types II and IV*, IV and II*, III and III* contribute identically. This was checked on the working set (`step9_twist_degree.py`): for the 157,938 pairs (curve, prime of type I_n^*) and the 292,580 pairs (curve, prime of potentially good additive type), $v_\ell(\deg \varphi_E) \geq v_\ell(\deg \varphi_{E_q})$ for $\ell \in \{3, 5, 7, 11, 13\}$, $\ell^2 \nmid N$, with no exception, and with equality in every pair of case (c), the strict inequalities occurring only for type I_0^* (case (b)), where the difference is the Euler term of Theorem 7.1.

Theorem 3.2. *Let $\ell \geq 5$, $\ell \nmid N$, with $\bar{\rho}_{E, \ell}$ absolutely irreducible on $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$. Let M be the set of primes q at which E has type I_n or (for q odd) I_n^* with $\ell \mid n$, and $M' \subset M$ the subset of primes $q \not\equiv \pm 1 \pmod{\ell}$. Then $v_\ell(\deg \varphi_E) \geq \sum_{q \in M'} v_\ell(n_q)$.*

Proof. Let S be the set of I_n^* primes in M' and E_S the twist of case (a). E_S is multiplicative at every prime of M' with the same n_q (Lemma 2.4 of the note), has conductor prime to ℓ , and $\bar{\rho}_{E_S, \ell} = \bar{\rho}_{E, \ell} \otimes \chi_S$ satisfies (TW). Apply (4) to f_{E_S} with N^- the product of the primes of M' (square-free, prime to N^+ , and (6) holds vacuously since $q \not\equiv \pm 1 \pmod{\ell}$): $v_\ell(r_{E_S}) \geq \sum_{q \in M'} v_\ell(n_q)$. Then $r_{E_S} \mid r_E$ and $v_\ell(\deg \varphi_E) = v_\ell(r_E)$. $\square$

This is Theorem 1.2 of the note extended to the twisted types; it proves the Tamagawa part of Conjecture 1.3 of the note in the Kim–Ota range. The Euler-factor part is the subject of the rest of this memorandum.

4. THE EULER FACTORS AT THE BAD PRIMES

Let $q \geq 5$, $q \neq \ell$, so that the action of I_q on V is tame, through a finite cyclic quotient for potentially good reduction and through a quadratic character times a unipotent for potentially multiplicative reduction. The representation $\mathrm{Sym}^2 V = \mathrm{ad}^0 V \otimes \chi_\ell$ has I_q -invariants and Frobenius eigenvalues on them as follows; for potentially good reduction $P_q(X)$ is the characteristic polynomial of the arithmetic Frobenius on the invariants of $\mathrm{Sym}^2 V$, in the normalisation of Section 1, and $P_q(q^{-2})$ is its value at $X = q^{-2}$.

Multiplicative, I_n . The local component of f is a Steinberg representation (up to an unramified quadratic twist), whose symmetric square has local factor $L_q(\mathrm{Sym}^2 f, s) = (1 - q^{-s})^{-1}$ in the classical normalisation (Shimura, Gelbart–Jacquet); this is also the factor $(1 - a_q^2 q^{-s})^{-1}$ of the naive Euler product, since $a_q^2 = 1$. Hence $e_\ell(E, q) = 0$: a multiplicative prime contributes only through its Tamagawa exponent, as the data confirm.

I_n^ .* $V = V' \otimes \chi_q$ with V' the Steinberg-type representation of the twist, and $\mathrm{Sym}^2 V = \mathrm{Sym}^2 V'$ since $\chi_q^2 = 1$; $a_q = 0$, so the naive factor is 1 while the primitive factor is that of the twist, $L_q(\mathrm{Sym}^2 f, s) = (1 - q^{-s})^{-1}$, $P_q(q^{-2}) = 1 - q^{-2}$, and $e_\ell = v_\ell(q^2 - 1)$.

II, II, IV, IV*.* The image of I_q in $\mathrm{Aut}(V)$ is cyclic of order 3 or 6, generated by σ or $-\sigma$ with σ of order 3 and eigenvalues $\zeta_3^{\pm 1}$. On $\mathrm{ad}^0 V$ the sign disappears and σ has eigenvalues $1, \zeta_3, \zeta_3^{-1}$: the invariants form a line, spanned by the element $\sqrt{-3} \in \mathbb{Z}_\ell[\zeta_3] \simeq \mathrm{End}_{I_q}(V) \cap \mathrm{ad}^0 V$. Frobenius normalises the image of inertia, $\mathrm{Frob} \sigma \mathrm{Frob}^{-1} = \sigma^q$, so it centralises σ when $q \equiv 1 \pmod{3}$ and inverts it when $q \equiv 2 \pmod{3}$; accordingly it acts on the invariant line by $+1$, resp. by -1 (conjugation sends $\sqrt{-3}$ to $-\sqrt{-3}$), that is, by $\chi_{-3}(q)$. On $\mathrm{Sym}^2 V$ the eigenvalue is $\chi_{-3}(q)q$, $P_q(q^{-2}) = 1 - \chi_{-3}(q)q^{-1}$, and $e_\ell = v_\ell(q - \chi_{-3}(q))$. Note $3 \mid q - \chi_{-3}(q)$ for every $q \geq 5$.

III, III.* Inertia acts through a cyclic group of order 4 with eigenvalues $\pm i$; the invariants of $\mathrm{ad}^0 V$ form the line of the diagonal element, on which Frobenius acts by $\chi_{-4}(q)$ (it centralises the generator when $q \equiv 1 \pmod{4}$ and inverts it otherwise). Hence $e_\ell = v_\ell(q - \chi_{-4}(q))$.

I_0^ .* $V = V' \otimes \chi_q$ with V' unramified at q , the representation of the good twist E' , with Frobenius eigenvalues $\alpha, \beta, \alpha + \beta = a_q(E')$, $\alpha\beta = q$. $\mathrm{ad}^0 V = \mathrm{ad}^0 V'$ is unramified, with Frobenius eigenvalues $\alpha/\beta, 1, \beta/\alpha$, so on $\mathrm{Sym}^2 V$ they are α^2, q, β^2 and

$$P_q(q^{-2}) = (1 - \alpha^2 q^{-2})(1 - q^{-1})(1 - \beta^2 q^{-2}) = \frac{q-1}{q} \cdot \frac{(q+1)^2 - a_q(E')^2}{q^2},$$

whence $e_\ell = v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$. This is Wiles's level-raising factor $(q-1)(a_q^2 - (q+1)^2)$ for the good form $f_{E'}$ at the prime q .

The prime ℓ itself, and $q = 2, 3$. Excluded: the inertia action is wild for potentially good reduction at 2 and 3, the local factors depend on more than the Kodaira type, and $q = \ell$ is outside the hypotheses.

5. THE DATA

All computations are on the working set (conductor ≤ 300000), hold-out ($300000 < N \leq 400000$), extension ($400000 < N \leq 500000$) and the 893 out-of-table curves of the note, with E optimal, non-CM and $E[\ell]$ irreducible. The scripts are the files `step9_*.py` of the repository and their outputs are in `results/step9_*`.

5.1. Each term in isolation. For curves with exactly one additive prime $q \geq 5$, $q \neq \ell$, and no multiplicative prime with $\ell \mid n$, Table 1 gives the minimum of $v_\ell(\deg \varphi_E)$ by type and by the value of $e_\ell(E, q)$. In every cell, for $\ell = 3, 5, 7$, the minimum equals e_ℓ and the proportion of curves with $v_\ell(\deg \varphi_E) \geq e_\ell$ is 1. For I_n^* the excess over e_ℓ has minimum exactly $v_\ell(n)$ in every cell $(e_\ell, v_\ell(n))$. No multiplicative prime shows an Euler term: for curves with a single multiplicative prime q with

$\ell \mid n$ and no additive prime, the minimum of $v_\ell(\deg \varphi_E) - v_\ell(n)$ is 0 for every value of $v_\ell(q^2 - 1)$ up to 6; with $\ell \nmid n$ the minimum of $v_\ell(\deg \varphi_E)$ is 0.

type ($\ell = 3$)	$e = 1$		$e = 2$		$e = 3$		$e = 4$	
	min	n	min	n	min	n	min	n
II	1	8,327	2	821	3	59	4	2
II*	1	4,184	2	296	3	12	4	1
IV	1	4,290	2	296	3	12	4	1
IV*	1	8,221	2	821	3	59	4	2
III	1	1,271	2	76	3	9	–	–
III*	1	1,271	2	76	3	9	–	–
I ₀ *	1	1,370	2	4,432	3	2,737	4	247
I _n *	1	30,497	2	2,589	3	145	4	13

TABLE 1. Working set, $\ell = 3$: minimum of $v_3(\deg \varphi_E)$ (and number of curves) over curves whose only additive prime is a single $q \geq 5$ of the given type with $e_3(E, q) = e$, and with no multiplicative 3-contribution. The cells with $e = 0$ (III: 5,534 curves, I₀*: 5,394) have minimum 0. For I₀* the term runs up to $e = 6$ (minimum 6, one curve). The same holds at $\ell = 5$ and $\ell = 7$ (files `step9_euler_15.txt`, 17), e.g. at $\ell = 5$ the cells $e = 1, 2$ of type I₀* have 4,876 and 999 curves with minima 1 and 2.

5.2. The rule on whole ranges. Table 2 checks (2) with all bad primes together, the wild primes 2 and 3 entering through the weights of the note (one for IV, IV* at 2 or 3 and for II, II* at 3, $v_3(n)$ for I_n* at 3). It also gives the rate of equality, which rises from 39% to 48% at $\ell = 3$ relative to the twist rule of the note, and the number of curves where the new bound is strictly larger.

5.3. The type III anomaly. All 264 violations at $\ell = 3$ of the rule with the III terms have the same shape: a prime of type III or III* with $e_3 \geq 1$ (so $q \equiv \pm 1 \pmod{12}$) together with a prime $p \equiv -1 \pmod{3}$ of type I_n or I_n* with $3 \mid n_p$ (in 258 cases a multiplicative one, in 6 cases one of type I_n*), and $v_3(\deg \varphi_E)$ falls short by exactly one. Dropping the III terms altogether leaves no violation on any range. For curves with a single additive prime $q \geq 5$ with $e_\ell \geq 1$ and some Tamagawa term, the sum $e_\ell + \sum t_\ell$ is a lower bound in every case for the types II, II*, IV, IV*, I₀, I_n (54,782 curves at $\ell = 3$, including 14,000 whose Tamagawa primes are all $\equiv -1 \pmod{3}$), while for type III it fails for 4.8% of the curves whose Tamagawa primes include one $\equiv -1 \pmod{3}$ and never when they are all $\equiv +1 \pmod{3}$ or $\not\equiv \pm 1$. Type III is the only type in the list at which f is minimal with $\bar{\rho}|_{I_q}$ semisimple and non-scalar (image of order 4, no fixed vector mod 3); the other types are either non-minimal (II, IV, I_n, I₀* after twist) or have $\text{ad}^0 \bar{\rho}$ unipotent at q . At $\ell = 5$ the analogous configurations (4,181 curves) show no failure.

5.4. The wild primes. For a single additive prime at $q \in \{2, 3\}$ and no multiplicative ℓ -part, the minima of $v_\ell(\deg \varphi_E)$ on the working set are: at $\ell = 3$ and $q = 2$: 1 for IV, IV* (15,024 and 12,756 curves), 1 for I_n* with $9 \mid n$ (3,699), 0 for the other types; at $\ell = 3$ and $q = 3$: 1 for II, IV, III*, 2 for II*, IV*, $v_3(n)$ for I_n, 0 for III and I₀*; at $\ell = 5, 7$: $v_\ell(n)$ for I_n at 3 and 0 in every other case, including I_n* at 2 with $\ell \mid n$. The last fact says that the Tamagawa term of an I_n* prime is absent at $q = 2$, where the twist to multiplicative reduction is by a character of conductor 4 or 8 and the index n of the twist can differ from that of E .

set	ℓ	curves	(2) > 0	viol.	equality	twist rule > 0 / viol. / equality	viol. with III	cond. III
working set	3	1,233,729	785,936	0	47.9%	705,563 / 0 / 39.4%	200	2
	5	1,303,224	300,973	0	74.0%	262,531 / 0 / 71.1%	0	0
	7	1,307,753	160,626	0	83.4%	136,707 / 0 / 81.2%	0	0
	11	1,309,103	55,844	0	91.2%	47,040 / 0 / 89.8%	0	0
	13	1,308,945	32,997	0	92.2%	29,943 / 0 / 91.5%	0	0
hold-out	3	409,577	271,615	0	45.0%	242,691 / 0 / 36.2%	42	2
	5	428,251	104,633	0	72.5%	89,164 / 0 / 69.0%	0	0
	7	429,280	56,167	0	82.4%	46,767 / 0 / 80.2%	0	0
	11	429,502	19,766	0	90.5%	15,970 / 0 / 89.0%	0	0
	13	429,498	11,702	0	92.0%	10,280 / 0 / 91.2%	0	0
extension	3	348,227	235,434	0	44.0%	214,033 / 0 / 35.5%	22	2
	5	358,244	90,357	0	71.5%	78,650 / 0 / 68.4%	0	0
	7	358,717	48,919	0	82.0%	41,317 / 0 / 79.6%	0	0
	11	358,830	17,891	0	90.3%	14,975 / 0 / 89.0%	0	0
	13	358,816	10,634	0	91.8%	9,465 / 0 / 91.2%	0	0
out of table	3	893	814	0	44.6%	718 / 0 / 34.7%	0	0
	5	893	89	0	83.1%	7 / 0 / 71.4%	0	0
	7	893	52	0	90.4%	1 / 0 / 0.0%	0	0
	11	893	19	0	94.7%	0 / 0 / –	0	0
	13	893	12	0	100.0%	0 / 0 / –	0	0

TABLE 2. The Euler-factor rule (2) on the four ranges: curves with positive bound, violations, equality rate; the same for the twist rule of the note; and the violations when the III terms are dropped, or kept only when no multiplicative prime $\equiv -1 \pmod{\ell}$ carries a Tamagawa term.

6. THE COMPARISON TABLE

type at $q \geq 5$, exponent of N	weight in the note	local representa- tion	exact local (this memo)	factor	status
$I_n, 1$	$v_\ell(n)$	Steinberg $\otimes$ unr.	$t = v_\ell(n), e = 0$		theorem for $\ell \geq 5$ (Kim–Ota, ARS) under (TW), $q \not\equiv \pm 1$ or $\bar{\rho}$ ramified at q ; data: all ℓ , all q
$I_n^*, 2$	$v_\ell(n)$	$\chi_q \otimes$ Steinberg	$v_\ell(n) + v_\ell(q^2 - 1)$		$v_\ell(n)$: Theorem 3.2; $v_\ell(q^2 - 1)$ when $\ell \nmid n$: Theorem 7.1 below; the sum when $\ell \mid n$: data only
$IV, IV^*, 2$	$[\ell = 3]$	tame, order 3	$v_\ell(q - \chi_{-3}(q))$		qualitative part (3.1) is Theorem 1.4(a) of the note; quantitative: data only (Section 8)
$II, II^*, 2$	$[\ell = 3] \text{ (odd } q)$	tame, order 6	$v_\ell(q - \chi_{-3}(q))$		equal to the IV^*, IV case by Theorem 3.1(b)
$III, III^*, 2$	0	tame, order 4	$v_\ell(q - \chi_{-4}(q))$		data only; attained in isolation, but not additive with the Tamagawa term of a prime $\equiv -1 \pmod{\ell}$ with $\ell \mid n$ in 5% of such cases at $\ell = 3$
$I_0^*, 2$	0	$\chi_q \otimes$ unramified	$v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$		Theorem 7.1 below (Wiles’s level-raising factor)
$2, 3, \ell$	as in the note	wild	not computed		data only (Section 5)

7. WHAT IS PROVABLE NOW

Theorem 7.1. *Let ℓ be an odd prime not dividing N with $\bar{\rho} = \bar{\rho}_{E,\ell}$ absolutely irreducible on $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, and let $q \geq 5$ be a prime at which E has type I_0^* , or type I_n^* with $\ell \nmid n$. Let $E' = E \otimes \chi_q$ (good, resp. multiplicative at q). Then*

$$v_\ell(\deg \varphi_E) \geq v_\ell(r_{E'}) + e_\ell(E, q),$$

with $e_\ell(E, q) = v_\ell(q-1) + v_\ell((q+1)^2 - a_q(E')^2)$, resp. $v_\ell(q^2 - 1)$, provided the Hecke module of [1, Theorem 4.20] is free, as it is under the hypotheses of [4, Theorem 2.9] ($\ell \geq 5$ for the statement as published; see the remark).

Sketch. Let $f' = f_{E'}$, of level $N' = N/q^2$, resp. N/q , and let Σ be the set of primes dividing $N/N(\bar{\rho})$; it contains q , and in both cases f' is minimal at q ($\bar{\rho}$ is unramified at q in the first case, ramified of Steinberg type in the second since $\ell \nmid n$), so f' lies in T_Σ and in $T_{\Sigma \setminus \{q\}}$, and $N_\Sigma = N'q^{\delta_q} \cdot (\dots)$ with $\delta_q = 2$, resp. 1, so that N_Σ is N up to the auxiliary prime. By (3) applied to f' in T_Σ and in $T_{\Sigma \setminus \{q\}}$, $\eta_{f',\Sigma} = \eta_{f',\Sigma \setminus \{q\}} \cdot c_q(f')$ with $c_q(f') = q^{\delta_q} L_q(\text{Sym}^2 f', 2)^{-1}$, whose valuation is $e_\ell(E, q)$ by Section 4. By [4, Lemma 2.7], $\eta_{f',\Sigma}$ is the congruence ideal of the q -depleted form $f'^{(q)}$ in the full Hecke algebra of level N (localised at $\bar{\rho}$), and $\eta_{f',\Sigma \setminus \{q\}}$ is the ℓ -part of $r_{E'}$. Finally $f = f'^{(q)} \otimes \chi_q$, and the twist map of Theorem 3.1 sends integral forms orthogonal to $f'^{(q)}$ at level N to integral forms orthogonal to f at level N , so the congruence number of $f'^{(q)}$ divides r_E ; and $v_\ell(\deg \varphi_E) = v_\ell(r_E)$ since $\ell \nmid N$. $\square$

Remark 7.2. (1) The statement is the quantitative form of the twist-and-lower argument of the note: the level is raised rather than lowered, from N' to $N'q^{\delta_q}$, and the price is Wiles’s Euler factor. (2) The identification of congruence ideals of depleted forms with Σ -ideals, [4, Lemma 2.7],

uses $\Sigma \supseteq P_{\bar{\rho}}$ and the freeness results of [3]; the published statements assume $\ell \geq 5$ (Kim–Ota’s main theorem) although their Assumption 2.1 only needs $k \leq \ell - 1$. For $\ell = 3$ the theorem should be regarded as conditional on the same freeness. (3) For I_n^* with $\ell \mid n$ the Tamagawa term is Theorem 3.2 and the Euler term is the same computation; their additivity is what the data assert and what (4) would give if it held for f' with $q \in N^-$ and (3) for the depleted form; the two are compatible only when (6) holds or the Euler factor is a unit, which is where the III anomaly also lives.

8. WHAT REMAINS OPEN, AND THE INTERPRETATION

For the non-minimal potentially good types (II, IV and their duals) the prime q is in Σ but f itself has level $N_f = N$, so (3) applied to f has no factor c_q : the Euler factor is inside the naive $L(\text{Sym}^2 f, 2)$, which omits the primitive factor at q . Writing L^{prim} for the primitive symmetric square,

$$\text{ord}_{\lambda} \eta_{f, \Sigma} = \text{ord}_{\lambda} \frac{N_f L^{\text{prim}}(\text{Sym}^2 f, 2)}{i\pi \det A} + \sum_{q \text{ additive}} e_{\ell}(E, q) + (\text{terms at } \ell),$$

so Conjecture 1.1 for these types is the statement that the primitive algebraic symmetric-square value is ℓ -integral, with the Tamagawa exponents of the multiplicative primes accounted for inside it (by the Bloch–Kato formula of [2] the primitive value is the order of the minimal adjoint Selmer group times the local Tamagawa factors of $\text{ad}^0 V$, and at a Steinberg prime that factor is $\ell^{v_{\ell}(n)}$). The same holds for the minimal type III with $q \notin \Sigma$. The anomaly of Section 5 says that the primitive algebraic value can have a denominator, by at most one factor of ℓ in the data, exactly in the situation (a Steinberg prime $p \equiv -1 \pmod{\ell}$ with $\bar{\rho}$ unramified at p , and a type III prime) where the Euler factor of $\text{ad}^0 V$ at p is a non-unit and Kim–Ota’s hypothesis (6) fails. A proof of Conjecture 1.1 for types II, IV, III therefore requires the Bloch–Kato formula for $\text{ad}^0 f$ with its local terms made explicit at the ramified primes, which is the content of [2] in the minimal case and of [4] at Steinberg primes; the type III anomaly is a prediction about the local Tamagawa factor at a Steinberg prime $p \equiv -1 \pmod{\ell}$ that the authors of [4] could presumably confirm or refute.

9. RECOMMENDATION FOR THE NOTE

- (1) Replace Conjectures 1.1 and 1.3 by Conjecture 1.1, stated for tame primes $q \geq 5$, $q \neq \ell$, with the wild primes and the type III caveat as separate statements. Present the component-group and twist-symmetrised forms as corollaries (since $3 \mid q - \chi_{-3}(q)$ and $3 \mid q^2 - 1$), which is how they were found.
- (2) Add Theorems 3.1, 3.2 and 7.1; keep Theorem 1.4 as the qualitative statement valid at $\ell = 3$ for types IV, II.
- (3) Put Section 8 in the introduction: the modular degree is governed by the naive symmetric-square L -value, and the local terms are the Euler factors that the naive value omits. This answers the referees’ question M1 and makes the type III anomaly the interesting open point.
- (4) Report the equality rates of Table 2 as the measure of sharpness, and the wild-prime minima as data.

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